

ORIGINAL ARTICLE

An enhanced experience-based gazelle optimization framework for mitigating partial shading condition problems in photovoltaic systems

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Abstract

Enhanced Gazelle Optimization Algorithm (EGOA) is proposed as an improved Gazelle Optimization Algorithm (GOA)-based framework to address the convergence-precision trade-off associated with fixed step-size search strategies. The proposed enhancement introduces a self-adaptive step-size mechanism that promotes rapid exploration during the early search stage and finer exploitation near the optimum. EGOA was evaluated against large-step GOA (LGOA), small-step GOA (SGOA), Grey Wolf Optimizer (GWO), Particle Swarm Optimization (PSO) and the conventional Perturb and Observe (P&O) algorithm using standard optimization benchmark functions and photovoltaic (PV) maximum power point tracking (MPPT) applications. In the optimization benchmarks, EGOA demonstrated a superior balance between convergence speed and solution accuracy compared with the fixed-step GOA variants and swarm-based baseline. In PV MPPT simulations, EGOA achieved exact MPPT under ideal irradiance and under all separated partial shading conditions (PSCs), where the average tracked power remained equal to the corresponding maximum power point (MPP) with near-zero standard deviation across repeated runs. By contrast, P&O exhibited larger deviations and significantly higher variability under the more challenging shading cases, while GWO and PSO showed intermediate performance. Under sequential step-change test, EGOA consistently recovered the global MPP more rapidly than the benchmark algorithms, with reduced oscillatory behavior. These results demonstrate that adaptive step-size regulation strengthens GOA for PV MPPT applications by improving convergence speed, optimization accuracy and tracking robustness under both static and sequential step-change of PSCs.

Keywords Optimization · Exploitation and exploration · Speed and accuracy · Partial shading conditions · Photovoltaic · Dynamic meta-heuristic

Abbreviations

AI	Artificial Intelligence
AOA	Arithmetic optimization algorithm
DC-DC	Direct current to direct current
EGOA	Enhanced gazelle optimization algorithm



GOA	Gazelle optimization algorithm
GWO	Grey wolf optimization
HC	Hill climbing
INC	Incremental conductance
LGOA	Large step size GOA
MPP	Maximum power point
MPPT	Maximum power point tracking
P&O	Perturb and observe
PMW	Pulse width modulation
PSCs	Partial shading conditions
PSO	Particle swarm optimization
PV	Photovoltaics
P-V	Power-voltage
SGOA	Small step size GOA
$P_{\max}$	Maximum power
G_n	Gain voltage
I_{mp}	Current at maximum power
I_{sc}	Short circuit current
P_0	Previous power measure
P_1	Current power measured
T_p	Time interval
V_{mp}	Voltage at maximum power
V_{oc}	Open circuit voltage
$X_a(r) Y_a(r)$	Search agent
$z_n(r)$	Optimal solution
Dd	Duty cycle
Δ	Search step size
a	Step size regulator
$f(z_{min})$	Minimum fitness function
j	Maximum iteration number
$x(i) y(i)$	Variables
$x_{minus} y_{minus}$	Subtraction of variables with step size
$x_{plus} y_{plus}$	Addition of variables with step size
$z(i) z_{Rxx} z_{Lxx} z_{Ryz} z_{Lyz}$	Optimized solution and its neighboring points

1 Introduction

Artificial intelligence (AI) and metaheuristic optimization algorithms are widely implemented in addressing complex engineering problems due to their ability to balance exploration and exploitation in global search processes [14, 17, 26]. In particular, population-based approaches have shown strong capability in solving nonlinear, multimodal and high-dimensional problems, thereby enhancing decision-making efficiency across various applications [19]. Within this domain, the GOA has emerged as a promising method, demonstrating superior performance compared with established algorithms such as the Arithmetic Optimization Algorithm (AOA), Particle Swarm Optimization (PSO) algorithm and Grey Wolf Optimization (GWO) [1, 2]. However, the conventional GOA relies on fixed step sizes, which restricts solution precision when steps are large and slows convergence when steps are small [12, 13].

In parallel, the demand of renewable energy has increased. Hence, MPPT has implemented to improve the efficient energy harvesting. Partial shading conditions (PSCs) create multiple local maxima in the power-voltage (P–V) curve, substantially reducing output efficiency [25, 30]. Conventional MPPT techniques such as Hill Climbing (HC), Incremental Conductance (INC) and Perturb and Observe (P&O) are widely adopted due to their simplicity, yet they suffer from oscillations, local maximum entrapment and reduced tracking accuracy under dynamic shading [3, 8]. Hence, metaheuristic optimization methods have been increasingly explored for MPPT, offering higher robustness under non-uniform irradiance [22].

Despite the large number of available optimization and bionic algorithms, refinement remains necessary because no single metaheuristic performs best across all problems and practical MPPT tasks under PSCs still demand a better compromise between convergence speed, accuracy and robustness [6, 7]. Large fixed step sizes accelerate movement toward promising regions but reduce precision near the optimum, whereas small fixed step sizes improve precision but substantially increase the iteration burden. Even advanced population-based methods, such as PSO and GWO, can still suffer from local entrapment or inconsistent convergence under rapidly varying shading [5, 16, 22]. Accordingly, this study does not claim an entirely different search paradigm, but it introduces an adaptive parameter-control enhancement to GOA that directly targets the fixed-step limitation, namely enhanced gazelle optimization algorithm (EGOA).

The main contributions of this study are fourfold. The contributions are presented as below,

- An adaptive step-size control layer is formulated for GOA, converting the fixed-step search into a self-regulated exploration–exploitation process.
- The relationship between fixed large-step, fixed small-step and adaptive-step GOA behavior is analyzed using eight benchmark functions to clarify the significance of enhancement.
- EGOA is benchmarked against representative optimizers, including LGOA, SGOA, GWO and PSO, using a common search budget.
- The enhanced framework is implemented for PV MPPT under ideal irradiance case, static PSCs and sequential step-change condition, where it is compared against the widely used P&O method and metaheuristic baselines.

In this paper, Sect. 2 explains the operational mechanism of the EGOA and its comparative performance across different step sizes. Additionally, it outlines the proposed EGOA framework and describes the methodology for the evaluation of all benchmarked algorithms. Section 3 presents the simulation results, highlighting key performance metrics such as solution accuracy, convergence behavior and search movement patterns. Finally, Sect. 4 concludes the research findings and proposes for the future development.

2 Enhanced gazelle optimization algorithm (EGOA)

2.1 Conventional design of gazelle optimization algorithm (GOA)

In their research, the researchers proposed the GOA, which replicates the survival strategies and behaviors exhibited by gazelles in their natural habitat [2]. Two primary phases are the dominant in this algorithm, which are the exploitation phase and the exploration phase. These phases work in tandem to approach optimal solutions by effectively addressing and optimizing complex problems.

As outlined in the pseudocode in Table 1, the fundamental concept of the algorithm involves comparing the solution obtained in each iteration with that of the previous one. The global optimum is obtained and updated. It will keep ongoing until it reaches the termination criterion. Then, the solution will be returned before the whole operation is terminated.

Table 1 Pseudocode developed by the researcher [2]

Pseudocode
Begin Initialization of parameter <i>s</i> limits from 0 to 1 μ limits from -1 to 1 <i>S</i> is defined as 88 kmph <i>PSRs</i> is defined as 0.45 <i>R</i> and <i>r</i> are randomized from 0 to 1 Initialization for search agents, the gazelle population while iter is less than the max_iter The fitness function was calculated for gazelles with the function Fittest gazelle = top—gazelle The matrix of elite gazelle is constructed if <i>r</i> is less than 0.5 exploitation: Gazelle is updated with $gazelle_{i+1} = gazelle_i + s.R * .R_B * .(Elite_i - R_B * .gazelle_i)$ Else exploitation: if iter divides by 2 is equal to 0 $\mu = -1$ else $\mu = 1$ if the number of rows for gazelle divided by 2 is less than iter for the gazelle populations ($i = 1, \dots, n/2$) update gazelles according to $gazelle_{i+1} = gazelle_i + S.\mu.R * .R_L * .(Elite_i - R_L * .gazelle_i)$ else for the gazelle populations ($i = 1, \dots, n/2$) update gazelles according to $gazelle_{i+1} = gazelle_i + S.\mu.R * .R_B * .(Elite_i - R_L * .gazelle_i)$ end if The fitness function is updated The top-gazelle is updated The elite gazelle is updated PSRs effect is applied to $gazelle_{i+1} = \begin{cases} gazelle_i + CF [LB + R * .(UB - LB)] * .U & \text{if } r \leq PSRs \\ gazelle_i + [PSRs (1 - r) + r] (gazelle_{r_1} - gazelle_{r_2}) & \text{else} \end{cases}$ end while return top-gazelle from the population End

2.2 Proposed enhanced gazelle optimization algorithm (EGOA)

EGOA is a global numerical optimization search algorithm derived from GOA. The hill climbing optimization algorithm serves as the primary inspiration for the development of EGOA, with the key distinction lying in the dynamic specification of the search step size. The EGOA developed according to the theoretical framework of a previously established algorithm [2]. The pseudocode for EGOA was formulated and accompanied by a detailed explanation of its operational functionality. The pseudocode was developed through program coding using MATLAB.

From Table 2, the notation ‘*j*’ represents the maximum iteration number employed for the entire search process, aimed at achieving the global optimum solution. In the simulation, this was initialized to 100 iterations. During the first iteration, the algorithm compares the values of test functions corresponding to the specific search

Table 2 Pseudocode proposed for EGOA

Pseudocode for the EGOA
initialize ten search agents, $X_a(r)$ and $Y_a(r)$, where $a = 1, 2, \dots, 10$
initialize the maximum iteration number, j
for $r=1$ to j do
if $r=1$
get the smallest $z_n(r)$ among the $z_n(r)$ of $X_a(r)$ and $Y_a(r)$, $a = 1, 2, \dots, 10$
adapt $x_n(r)$ as $x_{new}(r+1)$
adapt $y_n(r)$ as $y_{new}(r+1)$
adapt $z_n(r)$ as $z_{new}(r+1)$
else
get the smallest $z_n(r)$ among the $z_n(r)$ of $X_a(r)$ and $Y_a(r)$, $a = 1, 2, \dots, 10$ and $z_{new}(r)$
adapt $x_n(r)$ as $x_{new}(r+1)$
adapt $y_n(r)$ as $y_{new}(r+1)$
adapt $z_n(r)$ as $z_{new}(r+1)$
end if
end for

agents, denoted as $X_a(r)$ and $Y_a(r)$, to determine the optimal solution, referred to as $z_n(r)$. For subsequent iterations, the algorithm compares the newly obtained test function values from another set of search agents with the existed best optimal solution. It keeps processing until the algorithm meets the termination condition.

The step size and the number of seeds of EGOA is determined by testing with eight single-objective function for optimization. The average measurements are obtained from five times of simulation. The formula of performance for optimized result is calculated using Eq. (1), while the performance for time taken is calculated using Eq. (2). Hence, the best parameter setup for EGOA can be verified through the comparison with the baseline setting of 10 seeds with step size of 0.5 to 0.0005. Meanwhile, the equation used for the initial position of EGOA and the conventional GOA are same, which is shown as Eqs. (3) [2]. The initial position of the particular seeds is generated using the random build-in function in MATLAB.

$$\text{Performance (\%)} = \left(1 - \left| \frac{Z_{test} - Z_{opt}}{Z_{base} - Z_{opt}} \right| \right) \times 100\% \quad (1)$$

$$\text{Performance (\%)} = \frac{T_{test} - T_{base}}{T_{base}} \quad (2)$$

$$x_0, y_0 = \text{lower limit} + \text{rand} * (\text{upper limit} - \text{lower limit}) \quad (3)$$

Figure 1 shows the proposed EGOA is with implemented adaptive step size feature for the optimization. It shows that the variable a is employed to regulate the step size, while the search step size, Δ , is adjusted to the value determined through the testing in Sect. 3.1, whereas a is initialized by one and limited up to three. The following manipulated x and y are computed following Eqs. (4), (5), (6) and (7).

$$x_plus = x_i + \Delta, i = 0, 1, 2, \dots \quad (4)$$

$$y_plus = y_i + \Delta, i = 0, 1, 2, \dots \quad (5)$$

$$x_minus = x_i - \Delta, i = 0, 1, 2, \dots \quad (6)$$

$$y_minus = y_i - \Delta, i = 0, 1, 2, \dots \quad (7)$$

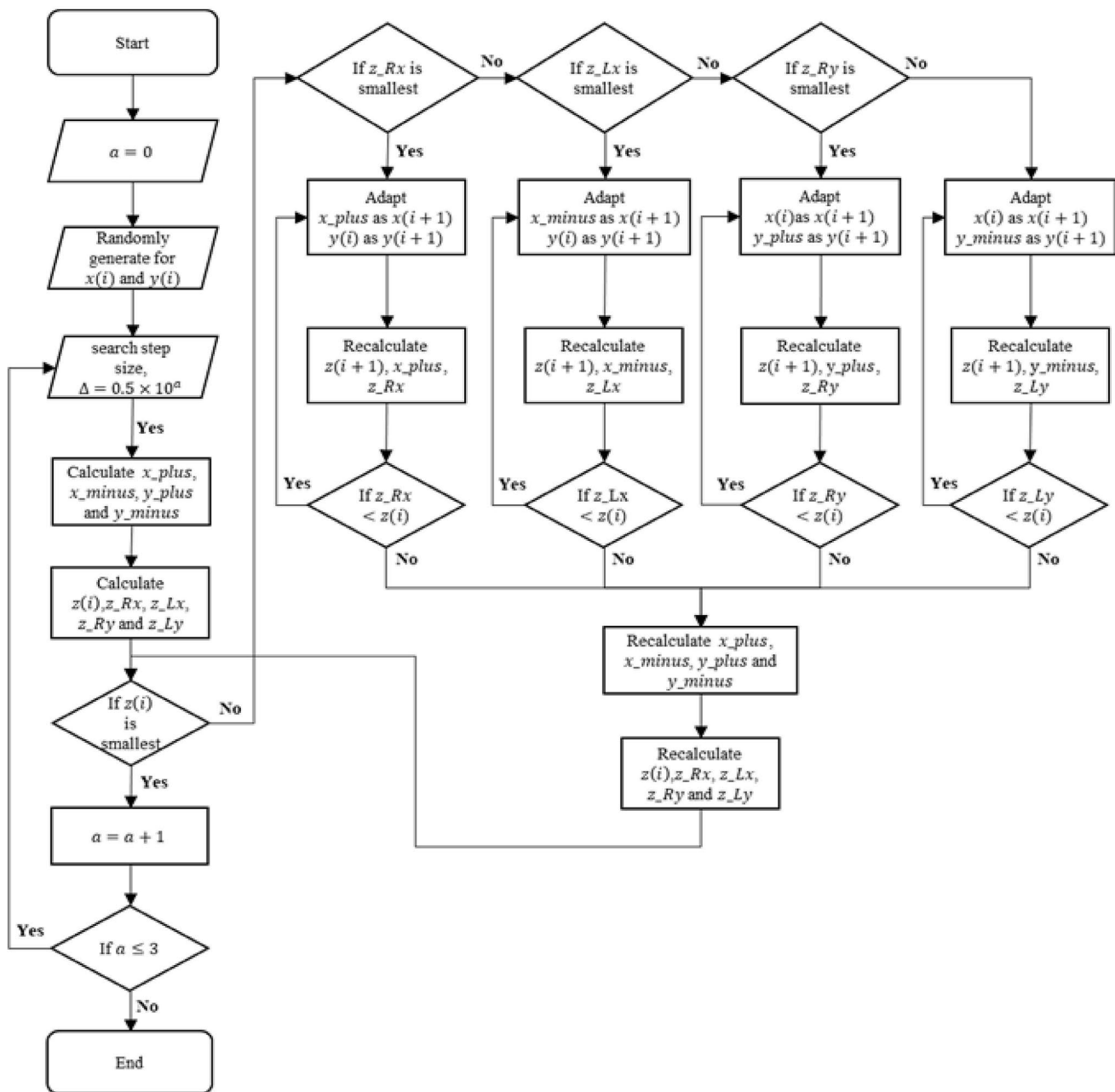


Fig. 1 Flowchart for the search agent of the EGOA

A larger step size facilitates a faster approach towards the optimal solution. The variables surrounded of initial position are determined by adding or subtracting the search step size, Δ . Subsequently, the test function values for $z(i)$, z_Rx , z_Lx , z_Ry and z_Ly are computed based on the combinations of variables. The values of $z(i)$, z_Rx , z_Lx , z_Ry and z_Ly are compared to determine the variables for the subsequent iteration, $x(i+1)$ and $y(i+1)$. The search will climb toward positive or negative of x or y , to obtain the smallest value of $z(i)$. Additionally, Δ is reduced by a factor of ten when a is less than and equal to three, provided that $z(i)$ is smaller than z_Rx , z_Lx , z_Ry and z_Ly . The simulation terminates once the stopping criteria are satisfied.

2.3 Testing suit setup for the EGOA

The comparison aimed to evaluate solution accuracy and convergence speed across these algorithms, which are EGOA, large step size GOA (LGOA), and small step size GOA (SGOA), GWO and PSO. The experimental methodology is expanded to improve transparency and facilitate independent replication. Specifically, Table 3 has been added, detailing the control parameters used for EGOA and the conventional GOA [15]. Furthermore, the manuscript now explicitly states that each algorithm was evaluated over 30 independent runs using randomly generated initial populations.

Regarding the random seeds, the simulations employed the default pseudo-random number generator of MATLAB, where seeds were generated automatically for each independent run to ensure statistical robustness and avoid bias associated with a single fixed seed. As the primary objective was comparative performance evaluation across multiple stochastic trials, the reported statistical metrics, including minimum, maximum, mean and standard deviation, provide a robust assessment of algorithm behavior. Nevertheless, the fixed random seeds would further enhance exact reproducibility. This has been identified as an important consideration for future work and the complete source code, including deterministic seed settings, will be made publicly available upon publication.

The absolute error and average absolute error will be computed to identify the accuracy of result optimized by algorithms using Eqs. (8) and (9).

$$\text{absolute error} = \sum |\text{optimized average value} - \text{true value}| \quad (8)$$

$$\text{average absolute error} = \frac{\text{absolute error}}{\text{number of datasets}} \quad (9)$$

Convergence analysis was conducted by plotting graphs of the convergence histories for each algorithm to monitor their local search behaviours over iterations. Additionally, scatter plots were created to visually represent the search movement history for the best-performing agent in each test function, offering insights into the decision variable convergence process for $x(i)$ and $y(i)$. Since GWO and PSO are retained as complementary metaheuristic baselines in the statistical and dynamic-search comparisons, the local-search conclusions are interpreted for EGOA against the conventional GOA in the eight-function benchmark and EGOA-versus-P&O comparison in the MPPT application, rather than as a complete ranking of all metaheuristics.

2.4 Test function for the optimization

The experimental simulations were performed on a 64-bit Windows 11 platform equipped with an Intel(R) Core (TM) i7-8550U processor running at 1.80 GHz and supported by 8 GB of RAM. MATLAB R2019a software was utilized to implement the simulation. The performance was evaluated by optimizing the test functions to obtain the most optimal solution, which, in this case, corresponds to the global minimum. Detailed information on the test functions used in the evaluation is provided in Table 4.

Table 3 Specification and parameter settings for the algorithms

Algorithms	Parameter settings
EGOA	$\Delta = 0.5 \times 10^a$, $a = 0, 1, 2, 3$
SGOA	$\Delta = 0.005$
LGOA	$\Delta = 0.5$

Table 4 Test function used for the optimization

Name	Equation	Boundaries	Global Minimum(x, y)
Ackley	$f(x, y) = -20 \exp \left[-0.2 \sqrt{0.5(x^2 + y^2)} \right] - \exp [0.5 (\cos 2\pi x + \cos 2\pi y)] + e + 20$	$x, y = [-5, 5]$	(0,0)
Beale	$f(x, y) = (1.5 - x + xy)^2 + (2.25 - x + xy^2)^2 + (2.625 - x + xy^3)^2$	$x, y = [-4.5, 4.5]$	(3,0.5)
Booth	$f(x, y) = (x + 2y - 7)^2 + (2x + y - 5)^2$	$x, y = [-10, 10]$	(1,3)
Goldstein-Price	$f(x, y) = [1 + (x + y + 1)^2 \times (19 - 14x + 3x^2 - 14y + 6xy + 3y^2)]$ $\quad \times [30 + (2x - 3y)^2 \times (18 - 32x + 12x^2 + 48y - 36xy + 27y^2)]$	$x, y = [-2, 2]$	(0, -1)
Lévi N.13	$f(x, y) = \sin^2(3\pi x) + (x - 1)^2 (1 + \sin^2(3\pi y)) + (y - 1)^2 (1 + \sin^2(2\pi y))$	$x, y = [-10, 10]$	(1,1)
Matyas	$f(x, y) = 0.26(x^2 + y^2) - 0.48xy$	$x, y = [-10, 10]$	(0,0)
Sphere	$f(x) = \sum_{i=1}^n x_i^2$	$x_i = [-\infty, \infty]$ $i = [1, \infty]$	(0, ..., 0)
Three-hump Cam	$f(x, y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2$	$x, y = [-5, 5]$	(0,0)

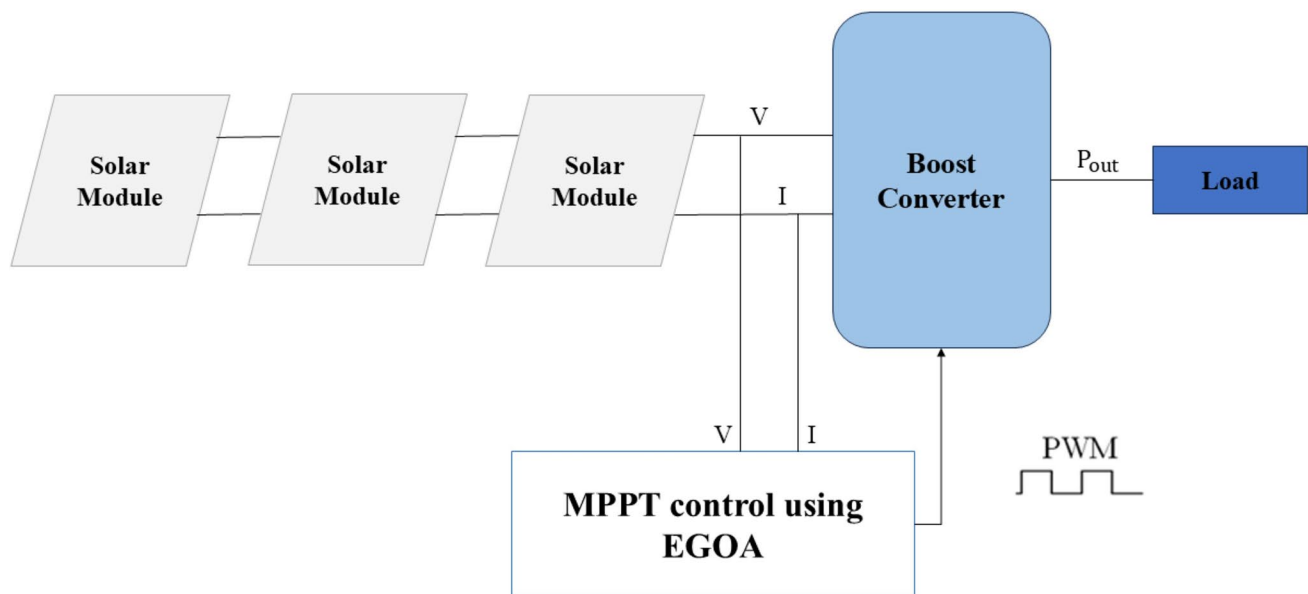


Fig. 2 System diagram designed for the solar MPPT mechanism simulation

Table 5 The main specifications of the LONGi Solar LR4-72HBD-415 M PV panel

Parameters	Value
Maximum Power, $P_{\max}$	445.00 W
Open Circuit Voltage, V_{oc}	90.00 V
Short Circuit Current, I_{sc}	6.24 A
Voltage at Maximum Power, V_{mp}	76.50 V
Current at Maximum power, I_{mp}	5.82 A

2.5 MPPT mechanism for PV arrays

The system configuration using the proposed EGOA is illustrated in Fig. 2. In this setup, three solar modules are in series connection and fed into a boost converter [18, 31]. The MPPT mechanism is implemented through a software package utilizing the EGOA, which continuously receives voltage and current readings from the solar modules. These real-time inputs are processed to generate pulse width modulation (PWM) signals, enabling precise manipulation of the boost converter to dynamically adjust the operating voltage and maximize power extraction for the load [21]. The boost converter regulates the voltage by modifying the duty cycle, denoted as ‘ d ’ in Eqs. (10) [28]. By applying this equation, the voltage gain, G_n , can be determined. Furthermore, the detailed electrical characteristics and operational parameters of the solar modules used in this system are summarized in Table 5.

$$G_n = \frac{1}{1 - d} \quad (10)$$

The simulation was initially conducted under ideal irradiance conditions. As outlined in Table 5, the P–V characteristics of the solar modules are depicted in Fig. 3. The maximum power point (MPP) is attained at 76.5 V, yielding an optimal power output of 1335 W. Given that the V_{oc} is 90 V, the simulated P–V curve extends from 0 to 90 V. Accordingly, the MPPT mechanism is designed to regulate the operating voltage near 76.5 V to enhance power extraction. Additionally, the P–V characteristics under PSCs are illustrated from Figs. 4, 5, 6, 7

The simulation of the proposed EGOA was conducted 30 times to reduce the effect of outlier to the dataset. Each run was independent, yielding 30 optimal solutions for each test function. Additionally, convergence

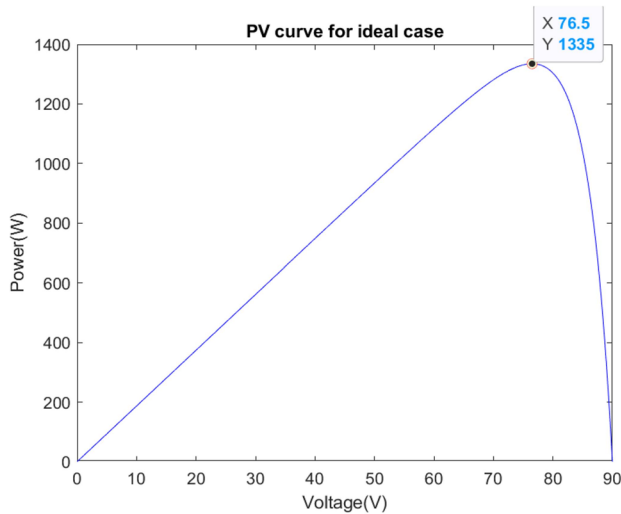


Fig. 3 P–V curve for the ideal irradiance case

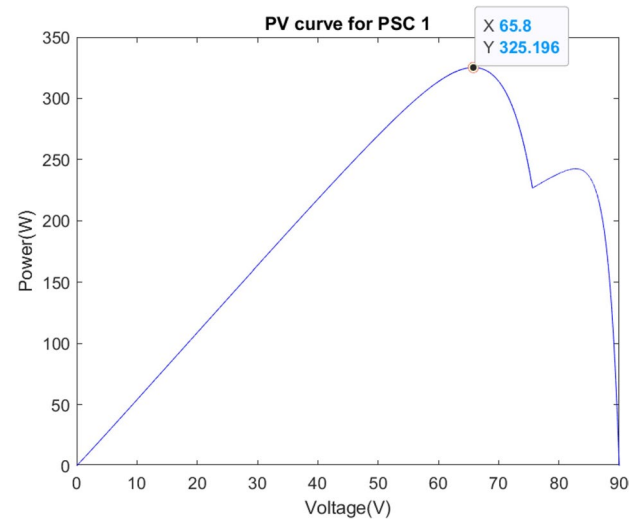


Fig. 4 P–V curve for PSC 1

histories were analyzed by observing the search iteration for each test function, and these were graphically represented. Furthermore, the movement history of the search agents was depicted using a scatter plot, which illustrates parts of the trajectory of the decision variables as they progressed toward the optimal solution.

Following the analysis of the proposed EGOA, the MPPT mechanism was re-simulated using the benchmarking algorithm currently employed in commercial applications, namely the P&O algorithm, together with the representative metaheuristic approaches, which are PSO and GWO [6, 7]. The overall results from the algorithms were then compared to identify the algorithm that performs the best in tracking MPP. The results illustrate the performance of algorithms in terms of efficiency, accuracy and convergence speed, allowing for a clear identification of the more effective MPPT method.

2.6 State-of-the-art MPPT framework

In this research, the tracking performance of MPPT algorithm in was benchmarked against the P&O algorithm. The algorithms work by modulating the duty cycle to optimize power tracking. The operational steps are illustrated in the flowchart provided in Fig. 8, detailing the process through which it regulates the d in response to changes in voltage and current to achieve MPPT [10, 34].

As observed from Fig. 8, d is adjusted to achieve the MPP. The algorithm applies a voltage variation at each time interval, T_p , and then compares the current power measured, P_1 , with the previous power measured, P_0 , for each d . The results of comparison decides the direction of the next perturbation, allowing the algorithm to adjust the duty cycle appropriately to optimize power tracking and converge toward the MPP [24].

Hence, the testing of algorithms, including EGOA, P&O, GWO and PSO, under independent PSCs and sequential step-change test will be conducted using the parameter settings illustrated in Table 6.

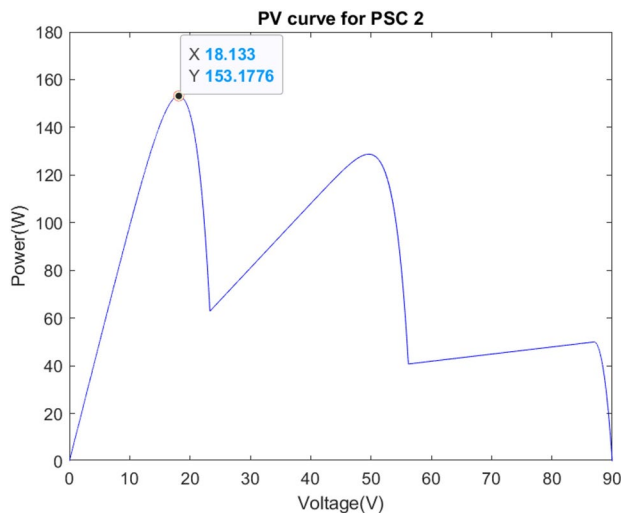


Fig. 5 P–V curve for PSC 2

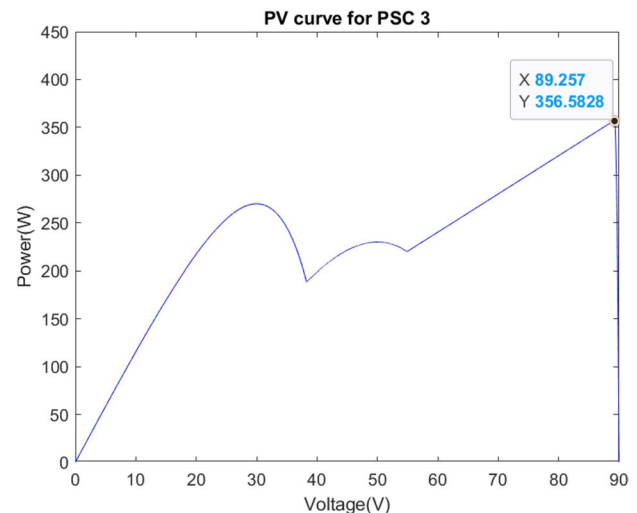


Fig. 6 P–V curve for PSC 3

3 Comparison and analysis of experimental results

3.1 Parameter settings for the proposed enhanced gazelle optimization algorithm (EGOA)

The parameter of the proposed EGOA is tuning through the testing using optimization test suite functions. The parameters included are number of seeds and the step size of algorithm. The number of seeds is examined from 9 to 11, while the step sizes are studied between two categories, which are from 0.5 to 0.0005 and 1 to 0.001. Table 7 shows the result of parameter tuning for EGOA under eight test suites single-objective function for optimization.

The parameter selection of EGOA was determined from the comparative results in Table 7, where the number of seeds and the adaptive step-size range were evaluated using eight single-objective benchmark functions. The result obtained were averaged over five independent simulation runs, while the relative performance of the optimized result and computational time was quantified using Eqs. (1) and (2), respectively. The results show that

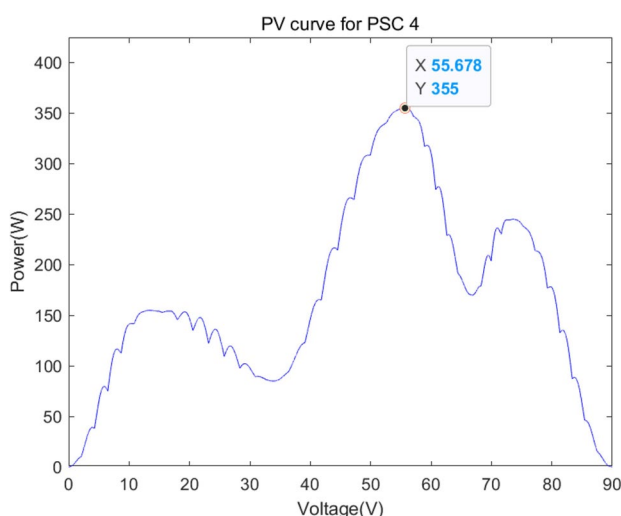


Fig. 7 P–V curve for PSC 4

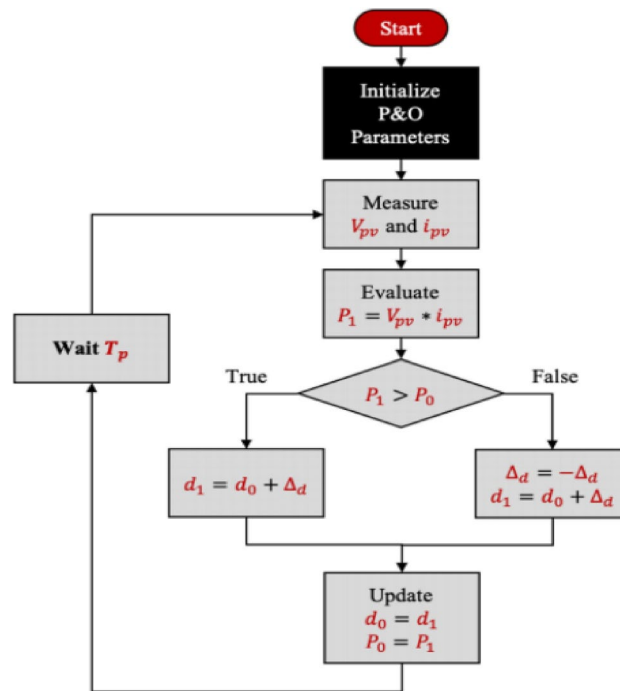


Fig. 8 Flowchart of the benchmarked MPPT mechanism [27]

the step-size range of $0.5 \sim 0.0005$ provides a more reliable balance between search accuracy and computational efficiency than the larger range of $1 \sim 0.001$. Across all tested seed counts, the $1 \sim 0.001$ setting did not produce the best objective value in any of the benchmark functions. In several cases, it resulted in substantially inferior optimization performance. This behavior can be attributed to an excessively large initial search step, which increases the tendency to overshoot promising regions, combined with a terminal step of 0.001 that remains relatively coarse for fine local refinement. In contrast, the $0.5 \sim 0.0005$ schedule preserves sufficient exploration capability during the early iterations, while providing an additional order of precision during the final exploitation stage. As a result, this range is better suited to the design objective of EGOA, namely to reconcile rapid global exploration with accurate convergence near the optimum.

With the step-size range fixed at $0.5 \sim 0.0005$, the comparison of 9, 10 and 11 seeds further reveals that 10 seeds offer the most balanced overall configuration. Although 9 seeds yield the lowest mean runtime, its objective values are weaker in several functions, particularly Ackley, Beale, Booth and Sphere, indicating that the reduced population diversity limits the robustness of the search process. On the other hand, increasing the population to 11 seeds improves the objective value in a subset of functions, such as Goldstein-Price, Lévi N.13, Matyas and Three-hump Camel, but this improvement is obtained at the expense of a clear computational penalty, with the

Table 6 Parameter setting for algorithm under various PSCs and sequential step-change test

Algorithms	Parameter settings
EGOA	$\Delta = 0.5 \times 10^a$, $a = 0, 1, 2, 3$
P&O	Δ_d is randomized between 0 to 0.5
GWO	a linearly decreased from 2 to 0
PSO	ω linearly decreased from 0.9 to 0.4, $c_1 = 2$, $c_2 = 2$

Table 7 Result of EGOA parameter tuning

Number of seeds		9			10			11		
Step size		0.5~0.0005			0.5~0.0005			0.5~0.0005		
Ackley $f(z_{min}) = 0$	$f(x, y)$	4.00E-05	3.95E-05	1~0.001	3.09E-05	4.34E-05	1~0.001	4.64E-05	1~0.001	4.54E-05
	Time taken (s)	0.214	0.215		0.241	0.226		0.255		0.246
	Performance $f(x, y)$	-29.41%	-27.80%		-	-40.40%		-50.03%		-46.74%
Beale $f(z_{min}) = 0$	Time taken	11.06%	10.56%		-	6.23%		-5.99%		-2.24%
	$f(x, y)$	1.12E-05	5.42E-06		9.60E-07	3.24E-06		2.32E-06		1.38E-06
	Time taken (s)	0.221	0.635		0.242	0.254		0.256		0.270
Booth $f(z_{min}) = 0$	Performance $f(x, y)$	-1066.67%	-464.58%		-	-237.50%		-141.67%		-43.75%
	Time taken	8.36%	-162.67%		-	-5.30%		-5.96%		-11.84%
	$f(x, y)$	2.35E-09	1.37E-08		1.73E-09	1.32E-08		2.14E-09		1.67E-08
Goldstein-Price $f(z_{min}) = 3$	Time taken (s)	0.215	0.217		0.231	0.229		0.249		0.243
	Performance $f(x, y)$	-36.00%	-694.26%		-	-664.87%		-23.61%		-867.88%
	Time taken	6.83%	5.97%		-	1.12%		-7.87%		-5.19%
Lévi N.13 $f(z_{min}) = 0$	$f(x, y)$	3.0000008315147	3.0000024888503		3.0000008800036	3.0000024401588		3.0000001772418		3.00000016018354
	Time taken (s)	2.05E-01	2.07E-01		2.17E-01	2.18E-01		2.32E-01		2.45E-01
	Performance $f(x, y)$	5.51%	-182.82%		-	-177.29%		79.86%		-82.03%
Matyas $f(z_{min}) = 0$	Time taken	5.26%	4.62%		-	-0.83%		-7.29%		-13.02%
	$f(x, y)$	2.36E-09	1.16E-08		2.33E-09	8.72E-09		1.89E-09		1.36E-08
	Time taken (s)	0.214	0.211		0.230	0.230		0.249		0.250
Sphere $f(z_{min}) = 0$	Performance $f(x, y)$	-1.10%	-399.10%		-	-274.39%		19.06%		-482.53%
	Time taken	6.71%	7.93%		-	-0.09%		-8.54%		-8.71%
	$f(x, y)$	4.39E-10	1.26E-09		4.37E-10	8.70E-10		2.83E-10		1.20E-09
Three-hump Camel $f(z_{min}) = 0$	Time taken (s)	0.203	0.210		0.219	0.226		0.230		0.247
	Performance $f(x, y)$	-0.45%	-187.41%		-	-98.89%		35.27%		-174.70%
	Time taken	7.31%	4.02%		-	-3.38%		-4.84%		-12.60%
	$f(x, y)$	1.73E-10	4.09E-10		3.82E-11	3.58E-10		8.41E-11		3.11E-10
	Time taken (s)	0.223	0.221		0.243	0.250		0.260		0.257
	Performance $f(x, y)$	-352.09%	-970.00%		-	-836.91%		-120.12%		-714.24%
	Time taken	8.15%	8.81%		-	-3.05%		-7.08%		-5.77%
	$f(x, y)$	7.00E-10	7.08E-10		6.40E-10	1.16E-09		2.69E-10		3.69E-10
	Time taken (s)	0.243	0.243		0.266	0.267		0.281		0.279
	Performance $f(x, y)$	-9.38%	-10.65%		-	-82.04%		57.96%		42.39%
	Time taken	8.71%	8.86%		-	-0.08%		-5.41%		-4.88%

Table 8 Result obtained by SGOA, EGOA, LGOA, GWO and PSO on each test function

Testing function	Parameter	Type of optimization algorithm				
		SGOA	EGOA	LGOA	GWO	PSO
Ackley $f(z_{min}) = 0$	Minimum	5.60E-04	4.51E-06	1.82E-02	3.20E-14	4.88E-07
	Maximum	4.06E-03	6.83E-05	3.33E-01	1.16E-03	6.83E-04
	Average	1.93E-03	3.44E-05	9.49E-02	8.42E-05	6.17E-05
	Std. Dev	9.66E-04	2.16E-05	6.08E-02	2.10E-04	1.27E-04
Beale $f(z_{min}) = 0$	Minimum	1.78E-07	9.86E-09	5.39E-05	5.84E-07	1.34E-13
	Maximum	7.19E-05	4.62E-06	8.79E-03	1.34E+00	7.62E-01
	Average	9.03E-06	1.55E-06	1.22E-03	1.68E-01	5.08E-02
	Std. Dev	1.38E-05	1.76E-06	1.60E-03	3.37E-01	1.90E-01
Booth $f(z_{min}) = 0$	Minimum	3.27E-06	1.04E-10	9.33E-05	6.08E-07	7.78E-14
	Maximum	1.22E-05	1.02E-08	2.42E-02	1.54E+00	1.86E-06
	Average	1.07E-05	3.56E-09	5.75E-03	1.14E-01	1.18E-07
	Std. Dev	2.30E-06	2.47E-09	5.97E-03	3.58E-01	4.22E-07
Goldstein-Price $f(z_{min}) = 3$	Minimum	3.000000094	3.000000001	3.008229621	3.000065169	3.000000000
	Maximum	3.000019217	3.000000140	3.571245738	91.119598793	3.000000676
	Average	3.000006263	3.000000038	3.158133720	9.216121435	3.000000058
	Std. Dev	5.11E-06	3.56E-08	1.35E-01	2.15E+01	1.65E-07
Lévi N.13 $f(z_{min}) = 0$	Minimum	4.27E-06	3.84E-11	5.03E-04	5.20E-07	2.17E-11
	Maximum	5.23E-04	1.16E-08	1.01E-02	2.47E-02	1.10E-01
	Average	1.47E-04	2.45E-09	4.89E-03	1.84E-03	4.03E-03
	Std. Dev	1.62E-04	2.32E-09	3.22E-03	4.64E-03	1.98E-02
Matyas $f(z_{min}) = 0$	Minimum	4.12E-08	2.88E-11	3.16E-06	4.14E-18	1.86E-13
	Maximum	1.20E-06	1.08E-09	7.16E-04	1.82E-08	3.19E-09
	Average	8.35E-07	2.30E-10	1.38E-04	1.03E-09	2.85E-10
	Std. Dev	4.46E-07	2.28E-10	1.56E-04	3.27E-09	6.83E-10
Sphere $f(z_{min}) = 0$	Minimum	4.71E-10	5.69E-13	1.21E-06	3.66E-29	1.00E-10
	Maximum	2.81E-08	2.65E-10	3.68E-04	4.02E-16	1.71E-03
	Average	8.13E-09	7.08E-11	9.88E-05	2.25E-17	1.36E-04
	Std. Dev	6.85E-09	7.60E-11	8.73E-05	7.62E-17	3.96E-04
Three-hump Camel $f(z_{min}) = 0$	Minimum	2.54E-11	4.39E-12	1.89E-05	3.32E-29	3.52E-13
	Maximum	1.65E-07	1.00E-09	9.38E-04	3.26E-15	2.99E-01
	Average	3.32E-08	3.30E-10	3.52E-04	1.16E-16	9.95E-03
	Std. Dev	4.00E-08	2.85E-10	2.77E-04	5.85E-16	5.36E-02

Table 9 Absolute error of the algorithms

Testing function	Type of optimization algorithm				
	SGOA	EGOA	LGOA	GWO	PSO
Ackley	1.93E-03	3.44E-05	9.49E-02	8.42E-05	6.17E-05
Beale	9.03E-06	1.55E-06	1.22E-03	1.68E-01	5.08E-02
Booth	1.07E-05	3.56E-09	5.75E-03	1.14E-01	1.18E-07
Goldstein-Price	6.26E-06	3.80E-08	1.58E-01	6.22E+00	5.80E-08
Lévi N.13	1.47E-04	2.45E-09	4.89E-03	1.84E-03	4.03E-03
Matyas	8.35E-07	2.30E-10	1.38E-04	1.03E-09	2.85E-10
Sphere	8.13E-09	7.08E-11	9.88E-05	2.25E-17	1.36E-04
Three-hump Camel	3.32E-08	3.30E-10	3.52E-04	1.16E-16	9.95E-03
Average	2.63E-04	4.50E-06	3.32E-02	8.13E-01	8.12E-03

average runtime increasing from 0.2364 s for 10 seeds to 0.2519 s for 11 seeds. More importantly, the 11-seed setting does not dominate the 10-seed setting consistently. Since the 10-seed configuration produces superior results in functions such as Ackley, Beale, Booth and Sphere, the selection of 10 seeds with a step-size range of 0.5~0.0005 is justified as the most appropriate trade-off between search diversity, convergence precision and execution time. Hence, this configuration avoids the insufficient exploration associated with 9 seeds and the

unnecessary computational burden introduced by 11 seeds, while maintaining consistently strong optimization performance across the tested benchmark suite.

3.2 Solution accuracies analysis

Table 8 illustrates the parameter values obtained across the test functions by each algorithm, offering a direct comparison of the EGOA, the conventional GOA with different fixed step sizes, GWO and PSO. The table includes key performance metrics such as the best, worst, mean, and standard deviation values for each test function, calculated over 30 independent simulation runs. This detailed comparison highlights performance differences in terms of solution accuracy and demonstrates the effectiveness of optimization achieved by the algorithm.

Table 8 shows that EGOA delivers the most balanced accuracy across the benchmark suite. Although GWO and PSO occasionally achieve extremely small minimum values on selected functions, such as Sphere and Three-hump Camel for GWO, and Beale and Booth for PSO, these isolated best cases are not sustained across all test functions. EGOA consistently returns average values that remain very close to the global minima for all eight benchmark functions, demonstrating its robustness even under adverse conditions [11]. Meanwhile, LGOA exhibits much larger average errors due to premature coarse-step convergence, while SGOA improves substantially over LGOA, but its mean values remain higher than those of EGOA on most functions. Table 9 further confirms this overall trend, which EGOA records the lowest average absolute error of 4.50×10^{-6} , outperforming SGOA (2.63×10^{-4}), PSO (8.12×10^{-3}), LGOA (3.32×10^{-2}), and GWO (8.13×10^{-1}). These results indicate that EGOA provides the strongest overall optimization accuracy among the compared algorithms (Gao et al. 2024).

However, accuracy alone does not fully establish algorithm consistency. The standard deviation values in Table 8 show that EGOA maintains small result dispersion across all eight functions, indicating highly repeatable convergence and high consistency over repeated runs [23, 36]. By comparison, LGOA exhibits noticeably larger variability because the large fixed step size increases the likelihood of overshooting the optimum. SGOA is generally more stable than LGOA, but it still shows wider dispersion than EGOA on several functions. GWO and PSO also reveal function-dependent instability, where GWO produces very large spread on Beale and Goldstein-Price, while PSO shows higher variability on Levi N.13, Sphere and Three-hump Camel. This behavior suggests that both swarm-based baselines can be highly competitive on selected landscapes, yet their performance is less uniformly reliable across the full benchmark set. Overall, the combination of low average absolute error and low standard deviation confirms that EGOA not only achieves accurate optimized solutions, but also preserves the most consistent overall behavior among the tested algorithms.

3.3 Convergence history analysis on global search

The global search convergence was evaluated over 30 independent runs, each comprising 100 iterations. The initial control variable (i_1) was set as the minimum value among 10 randomly generated seeds to establish a baseline. Figure 9 presents sampled global search convergence histories of SGOA, EGOA, LGOA, GWO, and PSO plotted on a logarithmic scale ($\log_{10}$) against iteration count, while Table 10 summarizes the optimized results and the corresponding iterations required for convergence. Then, these results enable direct comparison of the trade-off between convergence speed and final solution quality.

Table 10 and Fig. 9 reveal clear differences in the global search behavior of the five algorithms. LGOA often reaches a stable solution in the fewest iterations because its large fixed step size drives rapid early movement; however, this speed is frequently obtained at the expense of accuracy, as shown by the comparatively poor final results on all benchmark functions except for Booth function. SGOA improves the final solution quality relative to LGOA, but the smaller step size substantially increases the iteration requirement and can still lead to slower progress in multimodal landscapes [12, 13, 20]. GWO and PSO are capable of attaining highly accurate results on several individual functions, yet they generally require many more iterations to do so. This result is especially

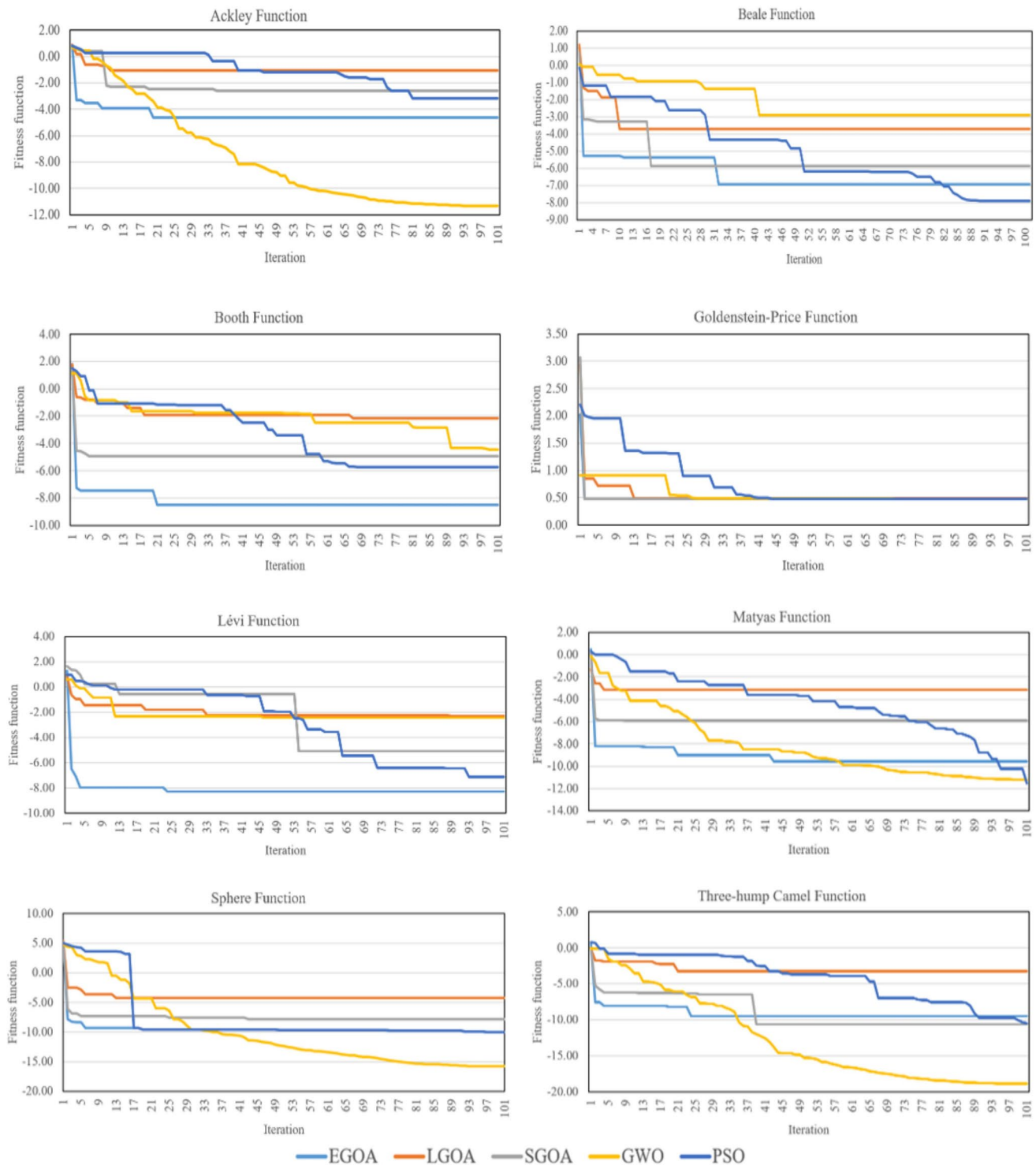


Fig. 9 The sampled global search convergence history of algorithms for the optimization test functions

Table 10 Optimized results and convergence iterations of SGOA, EGOA, LGOA, GWO and PSO

Testing function	Type of optimization algorithm									
	SGOA		EGOA		LGOA		GWO		PSO	
	Result	Ite	Result	Ite	Result	Ite	Result	Ite	Result	Ite
Ackley	2.47E-03	35	2.39E-05	20	8.46E-02	10	4.44E-12	99	6.83E-04	81
Beale	1.38E-06	17	1.19E-07	32	1.91E-04	10	1.27E-03	41	1.31E-08	90
Booth	1.14E-05	35	3.20E-09	21	7.19E-03	67	3.66E-05	98	1.86E-06	67
Goldstein-Price	3.000001	39	3.000000	22	3.110114	13	3.039121	85	3.000000	100
Lévi N.13	7.93E-06	54	4.93E-09	24	5.22E-03	89	3.54E-03	45	1.21E-02	92
Matyas	1.19E-06	73	2.74E-10	83	7.16E-04	14	6.42E-12	99	3.01E-12	100
Sphere	1.38E-08	96	1.00E-10	97	6.08E-05	13	1.58E-16	99	1.00E-10	96
Three-hump Camel	2.54E-11	39	3.14E-10	25	5.52E-04	21	1.38E-19	99	3.43E-11	100

evident for Ackley, Booth, Matyas, Sphere and Three-hump Camel, where GWO and PSO commonly approach their best solutions only near the end of the 100-iteration budget.

EGOA provides the most favorable overall compromise between convergence speed and final accuracy. It reaches competitive or superior solutions in relatively few iterations on many functions, including Ackley (20 iterations), Booth (21), Goldstein-Price (22), Levi N.13 (24) and Three-hump Camel (25), while maintaining much better final fitness values than LGOA and generally faster practical convergence than GWO and PSO. Although GWO achieves the smallest final values on functions such as Sphere and Three-hump Camel, and PSO performs strongly on Beale and Goldstein-Price, these gains are obtained only after markedly longer search durations. Therefore, the global convergence analysis indicates that EGOA is not necessarily the single fastest or most accurate method on every individual function [16], but it consistently delivers the strongest balance of convergence efficiency and solution quality across the full benchmark suite.

3.4 Convergence hisstory analysis on local search

Local search convergence was assessed using one representative seed from 30 simulation runs. The termination criterion was set to ensure each algorithm reached the optimal solution. Figure 10 presents the convergence curves of EGOA, LGOA, and SGOA across benchmark functions, with fitness values plotted on a logarithmic scale ($\log_{10}$) against iterations. This visualization enables a direct comparison of convergence behavior and performance among the algorithms.

Figure 10 shows that all algorithms converged towards smaller fitness values, confirming their optimization capability. SGOA, with a small step size of 0.005, demonstrated the slowest convergence and highest computational demand, though it achieved superior precision in the Beale function by avoiding overshooting of the global optimum [32]. LGOA, using a larger step size of 0.5, achieved rapid initial convergence but at the expense of accuracy, often terminating prematurely and yielding suboptimal solutions [4, 35]. In contrast, EGOA consistently balanced speed and precision through its adaptive step size ($0.5 \sim 0.0005$), enabling both fast early convergence and refined local search. Apart from the Beale function, EGOA outperformed both variants, with its convergence on the Sphere function closely resembling $\log_{10}(-x)$, confirming its effectiveness in approaching the optimum [9]. Overall, EGOA demonstrated superior convergence speed and solution accuracy, establishing it as the most effective optimization approach among the three.

3.5 Search agent movement analysis

The search agent movement was analyzed to examine the relationship between decision variables and fitness values across iterations. Figure 11 presents the trajectories of EGOA, LGOA, and SGOA, where blue, orange, and grey markers represent their respective search paths. These trajectories, sampled from randomly generated seeds, illustrate how each algorithm explores the solution space and converges towards the optimum.

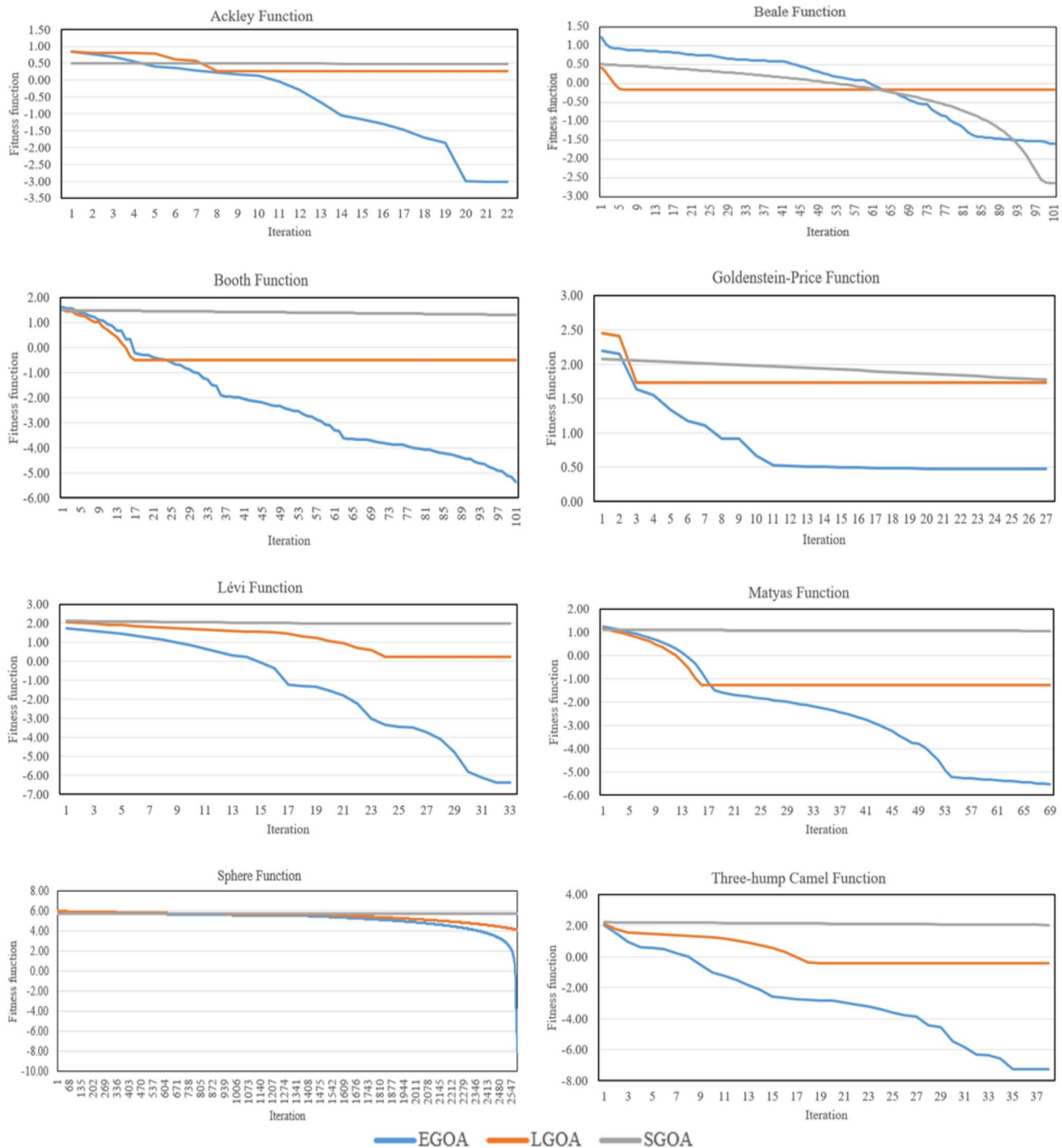


Fig. 10 The sampled local search convergence history of algorithms for the optimization test functions

Figure 11 illustrates the movement trajectories of LGO, SGO, and EGO during the optimization process. LGO and SGO display uniform spacing between consecutive points, directly reflecting their fixed step sizes of 0.5 and 0.005, respectively. This static behavior restricts flexibility, as LGO tends to halt prematurely with insufficient refinement, while SGO progresses slowly, demanding a larger number of iterations to achieve accuracy. The Beale function further highlights the search dynamics, where agents alternate between x- and y-axis adjustments to refine solutions. Such patterns demonstrate the capability of the algorithms to explore

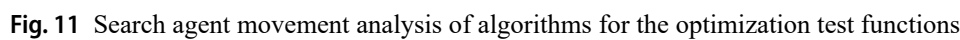


Table 11 Result obtained by algorithms on the different conditions

Type of testing function	Parameter	Type of optimization algorithm			
		EGOA (W)	P&O (W)	GWO (W)	PSO (W)
Ideal $f(z_{max}) = 1335W$	Minimum	1335.000	1334.954	1334.991	1334.973
	Maximum	1335.000	1335.000	1335.000	1335.000
	Average	1335.000	1334.996	1334.997	1334.989
	Std. Dev	0.00E+00	8.73E-03	2.11E-03	5.34E-03
PSC 1 $f(z_{max}) = 325.196W$	Minimum	325.196	325.031	325.192	325.164
	Maximum	325.196	325.196	325.196	325.196
	Average	325.196	325.183	325.195	325.185
	Std. Dev	1.22E-12	3.75E-02	1.19E-03	9.61E-03
PSC 2 $f(z_{max}) = 153.178W$	Minimum	153.178	131.400	153.160	151.622
	Maximum	153.178	153.178	153.178	153.178
	Average	153.178	150.999	153.173	152.541
	Std. Dev	6.82E-12	6.528	5.40E-03	2.39E-01
PSC 3 $f(z_{max}) = 356.583W$	Minimum	356.583	343.881	356.518	355.614
	Maximum	356.583	356.583	356.583	356.583
	Average	356.583	355.624	356.551	356.249
	Std. Dev	2.49E-11	2.963	8.63E-02	1.24E-01
PSC 4 $f(z_{max}) = 355W$	Minimum	355.000	244.990	354.729	353.661
	Maximum	355.000	355.000	355.000	355.000
	Average	355.000	351.325	354.904	354.283
	Std. Dev	3.24E-12	19.746	0.090	0.512

multi-dimensional landscapes but also emphasize the limitations of fixed step-size designs in adapting to varying optimization demands.

In contrast, EGOA exhibits non-uniform spacing across iterations, a result of its adaptive step-size strategy [36]. Larger steps enable rapid movement in the early search phase, while progressively smaller steps improve precision near the optimum. For instance, the wider gap observed between iterations i5 and i6, followed by a reduced gap at i7, demonstrates its ability to refine its trajectory dynamically. This adaptability allows EGOA to balance exploration and exploitation more effectively than its counterparts, ensuring both fast convergence and fine-grained optimization. Consequently, EGOA achieves superior efficiency, combining the speed of LGOA with the accuracy of SGOA, while avoiding their respective weaknesses.

3.6 Solution accuracies analysis

Table 11 presents the values of key parameters achieved by the EGOA, P&O, GWO and PSO algorithms under both ideal irradiance conditions and separated PSCs. These parameters include metrics related to maximum, minimum, average and standard deviation power, providing the illustration of performance of algorithm in optimizing MPPT under varying environmental conditions.

Table 11 further confirms the superior optimization accuracy of EGOA when the average tracked power is compared with the corresponding maximum power under each irradiance condition. Under ideal irradiance, the average power obtained by EGOA is 1335.000W, which is identical to the maximum value of 1335.000 W. A similar trend is observed under PSC1, PSC2, PSC3 and PSC4, where the average tracked powers of EGOA remain equal to their respective maximum values of 325.196W, 153.178W, 356.583W and 355.000W. This indicates that EGOA is able to converge to the global MPP with negligible mean error across repeated runs. By contrast, the average results of P&O deviate more noticeably from the maximum values, particularly under PSC2, PSC3 and PSC4, where the average tracked powers are 150.999W, 355.624W and 351.325W, respectively, compared with the corresponding maxima of 153.178W, 356.583W and 355.000W. These deviations suggest reduced tracking accuracy and a higher likelihood of entrapment in local maxima under more severe shading conditions. GWO and PSO produce average results that are generally closer to the maximum points than P&O, but they still

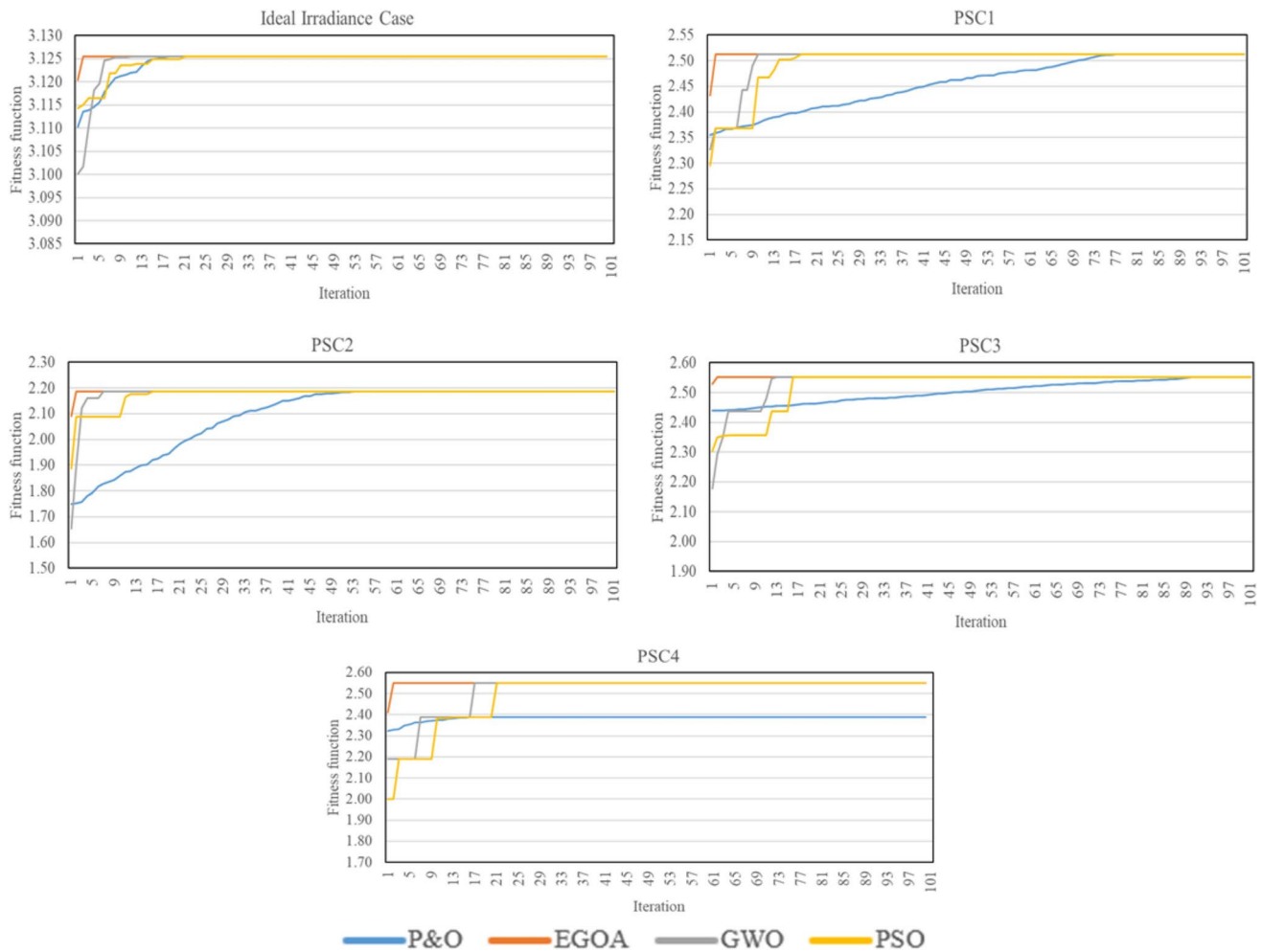


Fig. 12 The sampled global search convergence history under ideal irradiance case and PSCs

exhibit larger deviations than EGOA, especially under PSC3 and PSC4. Therefore, in terms of the closeness of the average tracked power to the maximum point, EGOA provides the most accurate optimization performance among the compared algorithms.

The standard deviation values in Table 11 further confirm the superior consistency of EGOA. Under ideal irradiance, EGOA records a standard deviation of zero, while under PSC1, PSC2, PSC3, and PSC4, the corresponding values remain extremely small at 1.22×10^{-12} , 6.82×10^{-12} , 2.49×10^{-11} and 3.24×10^{-12} , respectively. These near-zero values indicate that EGOA produces almost identical optimal solutions in every simulation run, demonstrating excellent robustness and repeatability across both mild and severe shading conditions. In comparison, P&O exhibits substantially larger standard deviations, namely 8.73×10^{-3} under ideal irradiance, 3.75×10^{-2} under PSC1, 6.528 under PSC2, 2.963 under PSC3 and 19.746 under PSC4, reflecting unstable tracking behavior and poor consistency when multiple local maxima are present. GWO and PSO show better consistency than P&O, but their standard deviations remain higher than those of EGOA, particularly under PSC3 and PSC4, where the optimization landscape becomes more challenging. Hence, the results in Table 11 indicate that EGOA not only yields the most accurate optimized solution through its average power values, but also the most consistent solution through its exceptionally low standard deviation [12, 13, 33].

3.7 Solutions exploration convergence analysis

Global search convergence was evaluated over 30 independent runs of EGOA and benchmarked against the P&O, GWO and PSO algorithm under both ideal irradiance and PSCs. Each run comprised 100 iterations, with the x-axis spanning from $i1$ to $i100$. Figure 12 presents the convergence curves of both algorithms, with fitness values plotted on a logarithmic scale ($\log_{10}$) against iteration count. The comparison highlights the relative convergence behavior and rates of algorithms under varying irradiance conditions.

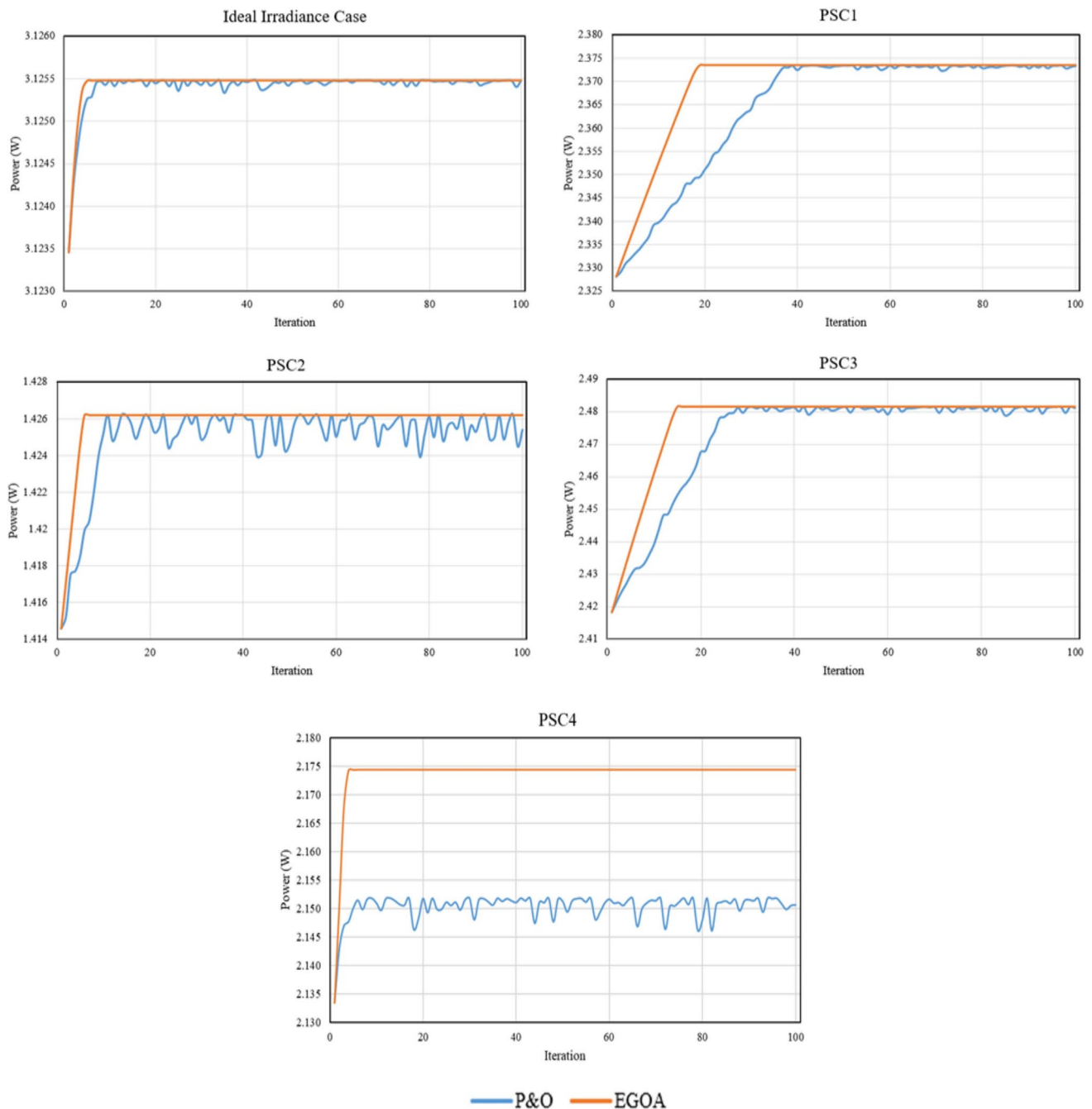


Fig. 13 Sampled local search convergence history under different irradiance cases

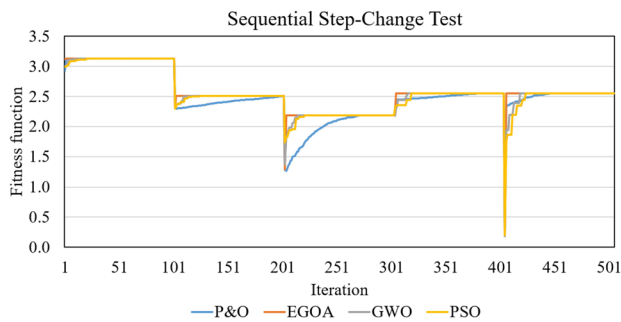


Fig. 14 The convergence history of the algorithms under the sequential step-change test

Figure 12 presents the sampled global search convergence histories of EGOA, P&O, GWO and PSO under ideal irradiance and PSCs, allowing direct comparison of the rate at which each algorithm approaches the optimized solution. Under ideal irradiance, all algorithms converge toward the global MPP, even clear differences in convergence speed can still be observed. EGOA shows the most rapid convergence behavior, reaching the neighborhood of the optimum within the earliest iterations. This trend remains evident under PSC1, PSC2, PSC3 and PSC4, where EGOA consistently demonstrates a steeper convergence trajectory than the other compared methods. The result suggests that the adaptive step-size mechanism enables effective global exploration at the start of the search while preserving sufficient refinement capability as the search approaches the MPP.

By comparison, P&O exhibits the slowest convergence behavior across all irradiance cases, and its progression becomes increasingly delayed as the shading condition becomes more challenging. This slower convergence is particularly apparent under PSC2, PSC3 and PSC4, where the search space contains more pronounced local maxima and the controller requires substantially more iterations to move toward the global optimum. GWO and PSO show intermediate convergence behavior, generally approaching the optimum faster than P&O but still more slowly than EGOA, especially under the separated PSC cases. The wider separation between the convergence curves under PSC4 further indicates that this case imposes a more difficult optimization landscape, making the advantage of adaptive step-size regulation more evident. Overall, the sampled global search convergence histories confirm that EGOA provides the strongest convergence speed among the compared algorithms under both ideal irradiance and increasingly complex PSCs.

3.8 Solutions exploitation convergence analysis

Local search convergence was assessed using one representative seed from 30 independent runs. The detailed exploitation-phase plot focuses on P&O because it is the classical MPPT reference most widely adopted in practice. Figure 13 has interpreted the local search convergence for EGOA and P&O, to study the behavior of each variable along the optimization process.

Figure 13 illustrates that EGOA and P&O exhibit comparatively similar local search behavior under ideal irradiance, as both are able to approach the MPP within the early stage of the iteration process. However, clear differences emerge once PSCs are introduced. Under PSC1, EGOA reaches the neighborhood of the MPP with fewer iterations than P&O, indicating a more efficient exploitation process after the initial search stage. Similar behavior is observed under PSC2 and PSC3, where EGOA maintains a steeper and more direct convergence path, while P&O progresses more slowly toward the optimum. This difference becomes even more evident under PSC4, where the local search landscape is more challenging and EGOA still demonstrates stable refinement toward the global MPP, whereas P&O is trapped by the local maximum.

The convergence histories of EGOA display a more structured progression toward optimality, enabled by the adaptive step-size mechanism that progressively reduces the step length as the operating point approaches the optimum. This allows the algorithm to preserve rapid early correction while improving local refinement near the MPP. In contrast, P&O exhibits less regular local search behavior, with slower progression and more visible oscillatory tendencies around high-power regions, particularly under PSC2, PSC3 and PSC4. Such behavior is consistent with the known limitation of conventional perturbation-based MPPT approaches, which may require more iterations to stabilize and can be more easily influenced by local maxima under PSCs [3], Choe, Tan, Mohammad Arif Sobhan, et al., 2023). Overall, the local search convergence results indicate that EGOA achieves faster and more stable exploitation than P&O under both uniform and non-uniform irradiance, with its advantage becoming increasingly pronounced as the shading pattern becomes more complex.

3.9 Convergence analysis under sequential step-change in PSCs

To evaluate the adaptability of the proposed method under time-varying operating conditions, EGOA was further assessed using a sequential step-change test that emulates discrete changes in irradiance patterns [29]. Although this protocol does not represent a fully dynamic irradiance profile, it provides a controlled means of examining algorithm behavior under abrupt transitions between shading states. Hence, the algorithm is simulated under a sequential step-change test, to observe the behavior of algorithms when they experience a sudden change. The test sequence began with ideal irradiance and then proceeded through PSC1, PSC2, PSC3 and PSC4, with each condition maintained for 100 iterations before the next transition was introduced. In this way, the operating voltage obtained at the end of one irradiance case was carried forward as the initial condition for the subsequent case. Figure 14 presents the convergence histories of EGOA, P&O, GWO and PSO under this step-change sequence, allowing direct comparison of their ability to recover the global MPP after each shading transition.

The convergence histories in Fig. 14 show distinct transition regions after every 100 iterations, corresponding to the change from one irradiance condition to the next. Across all transitions, EGOA demonstrates the fastest recovery toward the global MPP, indicating strong adaptability to abrupt changes in the power-voltage landscape. This behavior remains evident not only during the transitions from ideal irradiance to PSC1 and from PSC1 to PSC2, but also during the more challenging transitions involving PSC3 and PSC4, where multiple local maxima make the search process more difficult. In comparison, GWO and PSO exhibit slower recovery than EGOA, although both remain more responsive than P&O. Among the compared methods, P&O shows the weakest response, requiring a longer interval to approach the updated optimum after each transition and displaying more persistent delay under the more severe shading cases.

These results indicate that the adaptive step-size regulation of EGOA is effective not only in static PSC environments but also under a sequential step-change. By enabling rapid large-scale correction immediately after a transition and finer local refinement as the new optimum is approached, EGOA maintains a stable and efficient tracking behavior throughout the step-change sequence. The increasingly clear separation between EGOA and the benchmark algorithms in the later transition stages, particularly when PSC4 is introduced, further suggests that the proposed method retains its robustness even as the irradiance pattern becomes more complex. Overall, the sequential step-change analysis confirms that EGOA provides the most favorable balance of recovery speed and tracking stability among the compared algorithms under time-varying PSCs.

4 Limitation and future work

Although the proposed EGOA demonstrated excellent performance under ideal irradiance, static PSCs, and sequential step-change test, several limitations should be acknowledged. First, the dynamic shading conditions investigated in this study were based on sequential transitions among predefined PSC patterns. While these cases effectively evaluate adaptability, they do not fully capture highly irregular real-world shading phenomena, such

as rapidly moving clouds, transient obstructions from birds, or non-uniform contamination caused by dust and bird droppings. Such complex shading patterns may generate more intricate P–V characteristics with a larger number of local maxima, thereby increasing the difficulty of global MPP tracking. Second, the present validation was conducted entirely through MATLAB/Simulink simulations. Although simulation provides a reliable platform for preliminary evaluation, practical implementation may introduce additional challenges, including sensor noise, converter non-idealities, switching delays and hardware constraints.

Future research will therefore focus on two primary directions. The first is the development of more sophisticated shading models that closely emulate real-world harsh shading environments, including stochastic cloud movements, irregular shadow geometries and localized module contamination patterns. This will enable a more comprehensive assessment of the robustness of EGOA under practical operating conditions. The second is the implementation of hardware-in-the-loop and experimental prototype validation using real PV systems and embedded controllers. Such studies will allow the evaluation of computational burden, real-time responsiveness and tracking reliability under realistic environmental disturbances. These efforts will further strengthen the practical applicability of EGOA and facilitate its deployment in commercial PV energy systems.

5 Conclusion

The proposed Enhanced Gazelle Optimization Algorithm (EGOA) was developed as a GOA-based optimization framework with an adaptive step-size mechanism to overcome the limitations of fixed-step search strategies. The method was evaluated against LGOA, SGOA, GWO, PSO and the conventional P&O technique using standard optimization test functions as well as PV MPPT applications under ideal irradiance, separated PSCs and sequential step-change test. Across the benchmark functions, EGOA demonstrated an improved balance between convergence speed and solution precision, confirming that adaptive step-size regulation can enhance both global exploration and local exploitation relative to fixed large-step and fixed small-step variants.

In the PV MPPT application, EGOA consistently achieved the strongest overall performance under ideal irradiance and under PSC1, PSC2, PSC3 and PSC4. The algorithm maintained average tracked power values equal to the corresponding MPPs in all tested static irradiance cases, indicating highly accurate optimization performance. At the same time, the near-zero standard deviation obtained across all PSC cases confirmed excellent consistency and repeatability over repeated simulation runs. By comparison, P&O showed increasingly large deviations and substantially higher variability under the more challenging shading patterns, particularly in PSC2, PSC3 and PSC4, while GWO and PSO generally provided intermediate performance but remained less accurate and less stable than EGOA.

Under sequential step-change condition, EGOA also exhibited the fastest recovery toward the global MPP, indicating strong adaptability to abrupt changes in the P–V landscape. This observation suggests that the adaptive step-size mechanism is effective not only in static optimization tasks, but also in sequential tracking environments, where both rapid correction and precise refinement are required. Overall, within the tested simulation scenarios, EGOA provides a favorable combination of convergence speed, optimization accuracy and tracking robustness, making it a strong optimization framework for MPPT in PV systems. Future work should extend the present validation to more irregular real-world shading patterns and hardware-level implementation in order to further assess its practical applicability.

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Data availability The data that support the findings of this study are available from the corresponding author, J.D. Tan, upon reasonable request.

Declarations

Competing of interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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